ChatGPT said:

AI Course-II AIML/REC – Category 2

Practical Lab Tasks – 24/07/2026

Category 2 focuses on implementing intelligent agents in different AI environments using Python.

We'll use simple examples for each environment so you can easily understand and execute them in your practical lab.

Programs covered

No.	Environment	Python example
1	Fully Observable vs Partially Observable	Vacuum Cleaner Agent
2	Known vs Unknown	Environment Navigation Agent
3	Static vs Dynamic	Traffic Signal Agent
4	Discrete vs Continuous	Temperature Control Agent
5	Deterministic vs Stochastic	Robot Movement Agent
6	Episodic vs Sequential	Spam Classification and Chess Agent

1. Fully Observable vs Partially Observable Environment

Aim: To implement agents in fully observable and partially observable environments using a Vacuum Cleaner Agent.

Concept:

- Fully Observable: Agent knows the complete environment.
- Partially Observable: Agent has access to limited information.

Python Program

Python

```
# Fully Observable vs Partially Observable Agent

environment = {
    "A": "Dirty",
    "B": "Clean"
}

# Fully Observable Agent
def fully_observable_agent():
    print("FULLY OBSERVABLE")

    for room in environment:
        print("Room:", room)
        print("Status:", environment[room])

        if environment[room] == "Dirty":
            print("Cleaning Room", room)
            environment[room] = "Clean"

    print("Final Environment:", environment)


# Partially Observable Agent
def partially_observable_agent():

    print("\nPARTIALLY OBSERVABLE")

    current_room = "A"
```

Sample Output

Plain text

```
FULLY OBSERVABLE
Room: A
Status: Dirty
Cleaning Room A
Room: B
Status: Clean
Final Environment: {'A': 'Clean', 'B': 'Clean'}

PARTIALLY OBSERVABLE
Current Room: A
Agent cannot observe the other room.
```

Explanation:

1. Create two rooms, A and B.
2. Assign cleaning conditions.
3. In the fully observable environment, the agent checks both rooms.
4. In the partially observable environment, the agent checks only its current room.
5. Clean the room if it is dirty.

2. Known vs Unknown Environment

Aim: To implement an intelligent agent in known and unknown environments.

Concept:

- Known Environment: Agent knows the rules of the environment.
- Unknown Environment: Agent does not know the rules and learns through experience.

Python Program

Python

```
# Known vs Unknown Environment

# Known Environment
def known_environment():

    print("KNOWN ENVIRONMENT")

    environment = {
        "A": ["B", "C"],
        "B": ["D"],
        "C": ["D"],
        "D": []
    }

    print("Agent knows the environment.")

    current = "A"

    while current != "D":
        print("Current Location:", current)

        current = environment[current][0]

    print("Goal Reached:", current)


# Unknown Environment
def unknown_environment():
```

Sample Output

Plain text

```
KNOWN ENVIRONMENT
Agent knows the environment.
Current Location: A
Current Location: B
Goal Reached: D

UNKNOWN ENVIRONMENT
Exploring: A
Exploring: B
Goal Reached!
```

Explanation:

1. Create a graph representing locations.
2. In a known environment, the agent already knows the available paths.
3. In an unknown environment, the agent explores locations.
4. Store discovered paths in a learned environment.
5. Continue until the goal is reached.

3. Static vs Dynamic Environment

Aim: To implement agents in static and dynamic environments using a Traffic Signal example.

Concept:

- Static: Environment does not change while the agent is making a decision.
- Dynamic: Environment can change while the agent is making a decision.

Python Program

Python

```
# Static vs Dynamic Environment

# Static Environment
def static_environment():

    print("STATIC ENVIRONMENT")

    traffic = "RED"

    print("Traffic Signal:", traffic)

    if traffic == "RED":
        print("Agent Action: STOP")
    else:
        print("Agent Action: GO")


# Dynamic Environment
def dynamic_environment():

    print("\nDYNAMIC ENVIRONMENT")

    traffic_signals = ["RED", "GREEN", "YELLOW"]

    for traffic in traffic_signals:

        print("Traffic Signal:", traffic)

        if traffic == "RED":
```

Sample Output

Plain text

```
STATIC ENVIRONMENT
Traffic Signal: RED
Agent Action: STOP

DYNAMIC ENVIRONMENT
Traffic Signal: RED
Agent Action: STOP
Traffic Signal: GREEN
Agent Action: GO
Traffic Signal: YELLOW
Agent Action: SLOW DOWN
```

Explanation:

1. Create a traffic signal.
2. In the static environment, the signal remains unchanged.
3. In the dynamic environment, the signal changes.
4. The agent observes the signal.
5. The agent performs the appropriate action.

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4. Discrete vs Continuous Environment

Aim: To implement agents in discrete and continuous environments using temperature control.

Concept:

- Discrete: Environment has a limited number of states or values.
- Continuous: Environment can have a continuous range of values.

Python Program

Python

```python
# Discrete vs Continuous Environment

# Discrete Environment
def discrete_environment():

    print("DISCRETE ENVIRONMENT")

    temperature = 30

    if temperature > 25:
        action = "AC ON"
    else:
        action = "AC OFF"

    print("Temperature:", temperature)
    print("Action:", action)


# Continuous Environment
def continuous_environment():

    print("\nCONTINUOUS ENVIRONMENT")

    temperature = 28.5

    while temperature > 24:

        print("Temperature:", temperature)
```

Sample Output

Plain text

```
DISCRETE ENVIRONMENT
Temperature: 30
Action: AC ON

CONTINUOUS ENVIRONMENT
Temperature: 28.5
Temperature: 27.0
Temperature: 25.5
Target Temperature Reached!
```

Explanation:

1. Set a temperature value.
2. In the discrete environment, select an action based on a condition.
3. In the continuous environment, temperature changes gradually.
4. Continue until the target temperature is reached.

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5. Deterministic vs Stochastic Environment

Aim: To implement agents in deterministic and stochastic environments using robot movement.

Concept:

- Deterministic: The same action always produces the same result.
- Stochastic: The same action may produce different results due to randomness.

Python Program

Python

```python
import random

# Deterministic Environment
def deterministic_environment():

    print("DETERMINISTIC ENVIRONMENT")

    position = 0

    for i in range(3):
        position += 1
        print("Robot Position:", position)

    print("Final Position:", position)


# Stochastic Environment
def stochastic_environment():

    print("\nSTOCHASTIC ENVIRONMENT")

    position = 0

    for i in range(3):

        movement = random.choice([-1, 1])

        position += movement
```

Sample Output

Plain text

```
DETERMINISTIC ENVIRONMENT
Robot Position: 1
Robot Position: 2
Robot Position: 3
Final Position: 3
```

```
STOCHASTIC ENVIRONMENT
Robot Position: 1
Robot Position: 0
Robot Position: 1
Final Position: 1
```

Explanation:

1. Initialize the robot position.
2. In the deterministic environment, the robot always moves one step forward.
3. In the stochastic environment, movement is randomly selected.
4. Update the robot position.
5. Display the final position.

---

6. Episodic vs Sequential Environment

Aim: To implement agents in episodic and sequential environments.

Concept:

- Episodic: Each action is independent of previous actions.
- Sequential: Current actions affect future states and decisions.

Python Program

Python

```python
# Episodic Environment
def episodic_environment():

    print("EPISODIC ENVIRONMENT")

    tasks = ["Spam Detection", "Spam Detection", "Spam Detection"]

    for task in tasks:

        email = "Spam"

        if email == "Spam":
            print(task, "-> Classified as Spam")
        else:
            print(task, "-> Classified as Not Spam")


# Sequential Environment
def sequential_environment():

    print("\nSEQUENTIAL ENVIRONMENT")

    position = 0
    goal = 5

    while position < goal:
```

Sample Output

Plain text

```
EPISODIC ENVIRONMENT
Spam Detection -> Classified as Spam
Spam Detection -> Classified as Spam
Spam Detection -> Classified as Spam

SEQUENTIAL ENVIRONMENT
```

```
Current Position: 1
Current Position: 2
Current Position: 3
Current Position: 4
Current Position: 5
Goal Reached!
</CodeBlock>

**Explanation:**
1. In the episodic environment, each email is classified independently.
2. The result of one classification does not affect another.
3. In the sequential environment, the agent moves towards a goal.
4. Each movement changes the current position.
5. The agent continues until the goal is reached.

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## Important viva questions

| Question | Answer |
|---|---|
| What is a fully observable environment? | An environment where the agent can access the complete state. |
| What is a partially observable environment? | An environment where the agent has limited information. |
| What is a known environment? | An environment whose rules are known to the agent. |
```